Artificial Intelligence and Musculoskeletal Surgical Applications

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CLINICAL & AGENTIC AI TAKEAWAY

AI-assisted procedures in orthopedic surgery improve precision, efficiency, and patient outcomes while maintaining human clinical expertise.

ABSTRACT

Artificial intelligence (AI) has emerged as a transformative force in orthopedic surgery. Potentially encompassing pre-, intra-, and postoperative processes, it can process complex medical imaging, provide real-time surgical guidance, and analyze large datasets for outcome prediction and optimization. AI has shown improvements in surgical precision, efficiency, and patient outcomes across orthopedic subspecialties, and large language models and agentic AI systems are expanding AI utility beyond surgical applications into areas such as clinical documentation, patient education, and autonomous decision support. The successful implementation of AI in orthopedic surgery requires careful attention to validation, regulatory compliance, and healthcare system integration. As these technologies continue to advance, maintaining the balance between innovation and patient safety remains crucial, with the ultimate goal of achieving more personalized, efficient, and equitable healthcare delivery while preserving the essential role of human clinical judgment. This review examines the current landscape and future trajectory of AI applications in orthopedic surgery, highlighting both technological advances and their clinical impact. Studies have suggested that AI-assisted procedures achieve higher accuracy and better functional outcomes compared to conventional methods, while reducing operative times and complications. However, these technologies are designed to augment rather than replace clinical expertise, serving as sophisticated tools to enhance surgeons' capabilities and improve patient care. © The Author(s) 2025.

Keywords: Artificial intelligence, Spine, Knee, Hip, Machine Learning, Large Language Models, Agentic Ai

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Abstract

Artificial intelligence (AI) has emerged as a transformative force in orthopedic surgery. Potentially encompassing pre-, intra-, and postoperative processes, it can process complex medical imaging, provide real-time surgical guidance, and analyze large datasets for outcome prediction and optimization. AI has shown improvements in surgical precision, efficiency, and patient outcomes across orthopedic subspecialties, and large language models and agentic AI systems are expanding AI utility beyond surgical applications into areas such as clinical documentation, patient education, and autonomous decision support. The successful implementation of AI in orthopedic surgery requires careful attention to validation, regulatory compliance, and healthcare system integration. As these technologies continue to advance, maintaining the balance between innovation and patient safety remains crucial, with the ultimate goal of achieving more personalized, efficient, and equitable healthcare delivery while preserving the essential role of human clinical judgment. This review examines the current landscape and future trajectory of AI applications in orthopedic surgery, highlighting both technological advances and their clinical impact. Studies have suggested that AI-assisted procedures achieve higher accuracy and better functional outcomes compared to conventional methods, while reducing operative times and complications. However, these technologies are designed to augment rather than replace clinical expertise, serving as sophisticated tools to enhance surgeons' capabilities and improve patient care.

Keywords:artificial intelligence, machine learning, large language models, agentic AI, hip, knee, spine

Introduction

Artificial intelligence (AI) has emerged as a transformative force in healthcare, with considerable implications for orthopedic surgery. It has the potential to provide support at every phase of an increasingly complex surgical process, from preoperative planning to postoperative monitoring. The integration of machine learning (ML), computer vision, and advanced robotics is reshaping how orthopedic surgeons approach patient care, potentially leading to improved outcomes and enhanced surgical precision.

In total knee arthroplasty (TKA), AI-powered systems can enable automatic segmentation and 3-dimensional (3D) reconstruction of anatomical structures [28], while in total hip arthroplasty (THA), ML algorithms can aid in selecting patients and predicting outcomes [23,24]. Particular promise has been shown in reducing complications and improving surgical accuracy, for example, in enhancing component positioning and patient outcomes [18,45]. The emergence of large language models (LLMs) has further expanded AI utility in orthopedic surgery, facilitating ambient clinical documentation and patient education [3,10]. Meanwhile, agentic AI systems are advancing surgical autonomy and decision-making capabilities, while maintaining essential human oversight and ethical considerations [4,11,43].

This review explores the current state and future potential of AI applications in orthopedic surgery, including surgical planning, intraoperative navigation, and postoperative monitoring. We analyze the technical foundations of these advances, including innovative algorithms like U-Net architecture for surgical tool tracking [1] and you only look once (YOLO) implementations for real-time surgical phase recognition [14]. In addition, we examine the clinical evidence supporting these technologies and discuss the emerging roles of LLMs and agentic AI in advancing orthopedic surgical care.

Technical Foundations

"AI" is not one technology but rather is an umbrella term describing systems that display intelligent behavior by analyzing their environment to achieve predefined goals. These behaviors are defined by models, which result from an algorithm and its training environment. This section introduces some commonly utilized algorithms and their potential clinical use.

The U-Net architecture has emerged as a tool for surgical tool tracking during arthroscopic procedures. It is a fully convolutional neural network (CNN), known for its effectiveness in medical image segmentation tasks, which has been adapted for real-time instrument detection and segmentation in arthroscopic videos. U-Net's encoder-decoder structure with skip connections allows it to capture both fine-grained details and broader contextual information, making it particularly suitable for identifying surgical tools during knee surgery [1,15]. Recent studies have demonstrated U-Net's superior performance in segmenting visually challenging arthroscopic images, even with limited datasets, which is a common constraint in medical imaging research [1]. Modifications to the original U-Net architecture, such as incorporating pretrained encoders and replacing transposed convolutions with nearest-neighbor interpolation in the decoder, have further improved its accuracy and reduced artifacts in tool segmentation [16]. These enhancements have led to significant improvements in segmentation accuracy and computational efficiency, making U-Net-based approaches increasingly viable for real-time surgical tool tracking in arthroscopic procedures [16,21].

YOLO implementations have shown promising results in real-time surgical phase recognition, offering significant improvements in speed and accuracy compared to traditional methods. For instance, a modified YOLOv7 model, called YOLOv7-RepFPN, achieved an impressive 88.2% mean average precision at 62.9 frames per second on an embedded device, significantly outperforming the original YOLOv7 in terms of speed while maintaining comparable accuracy [22]. This enhancement in inference speed is crucial for real-time applications in the operating room. Similarly, an improved YOLOv5 algorithm has been proposed for surgical instrument recognition, incorporating advanced techniques such as the squeeze-and-excitation attention mechanism and distribution shifting convolution to enhance detection performance [14]. These YOLO implementations excel in speed and accuracy, and they demonstrate good generalization capabilities, making them suitable for the dynamic and complex environment of surgical procedures.

ResNet systems are deep learning (DL) models that have demonstrated remarkable efficacy in identifying and classifying various fracture types in orthopedic imaging. For ankle fractures, an adapted ResNet50 model enhanced with squeeze-and-excitation network capabilities achieved 93% accuracy, 95% area under the curve, and 92% recall [42]. In pelvic fracture classification, a combination of attention U-Net for segmentation and Inception-ResNet V2 for fracture categorization has shown significant improvements over traditional methods [19]. ResNet50 has also outperformed other neural networks in distinguishing atypical femoral fractures from normal femoral shaft fractures, achieving a mean accuracy of 91% in an automated pathway and 94% with manual intervention [48]. For identifying proximal humerus fractures, a ResNet152 model has achieved an outstanding accuracy of 96%, highlighting its effectiveness in emergency settings [17]. These results suggest ResNet systems' potential to enhance the accuracy and efficiency of fracture diagnosis in clinical practice.

Clinical Applications

With advancements in computational power and imaging technologies, AI applications are transforming preoperative planning, intraoperative guidance, and postoperative outcomes.

Surgical Planning

The use of AI for analyzing computed tomography (CT) and magnetic resonance imaging (MRI) has enabled automatic segmentation and precise 3D reconstruction of anatomical structures in TKA [20]. Traditional methods rely heavily on manual segmentation, which is labor-intensive and subject to inter-observer variability. AI algorithms, particularly those employing CNNs, have demonstrated high accuracy in segmenting bone and soft tissue structures, reducing the time required for preoperative planning. This improved efficiency allows for enhanced surgical precision by facilitating patient-specific implant design and alignment strategies. Studies have shown that automated segmentation tools can achieve comparable or superior accuracy to manual techniques while significantly reducing processing time [37].

In addition, ML has shown significant promise in optimizing various aspects of THA. Recent studies have demonstrated the potential of AI and ML models to aid in patient selection, predict postoperative outcomes, and analyze imaging data.

For instance, researchers have developed ML models to predict short lengths of stay and same-day discharge after THA, which could help optimize patient selection for outpatient procedures [23]. In addition, AI applications have shown the potential to accurately predict postoperative complications, pain, and patient-reported outcomes for THA [24]. AI using ML has been able to distinguish the final rasping hammering sound from prior hammering sounds during THA, which could help reduce complications by informing the surgeon that the cup or stem is fully seated [12]. Furthermore, ML methods have been explored for the surveillance of primary THA components and the early identification of outliers [9]. These advancements highlight the growing role of ML in enhancing various aspects of THA, from preoperative planning to postoperative monitoring.

AI-powered tools are advancing the evaluation of spinal alignment parameters critical for deformity correction surgery. Accurate assessment of sagittal and coronal alignment, pelvic tilt, and lumbar lordosis is essential for developing corrective strategies. DL algorithms trained on radiographic datasets can automatically identify key anatomical landmarks and calculate alignment parameters with high accuracy. Furthermore, predictive analytics assist surgeons in simulating postoperative outcomes and selecting optimal surgical interventions. Research indicates that these AI systems reduce variability and improve the reliability of spinal alignment assessments [29,50].

Intraoperative Navigation

Real-time AI processing of fluoroscopic images for pedicle screw placement has shown significant promise in enhancing surgical accuracy and efficiency. A novel AI-based fluoroscopy reconstruction technique called X23D generates a 3D anatomical model of the spine from only 4 fluoroscopy images, enabling real-time visualization of the spine anatomy and surgical drill position during pedicle screw placement [25]. This technology has demonstrated improved accuracy compared to traditional fluoroscopy-aided freehand instrumentation, with reduced radiation exposure and execution time.

AI-enhanced visualization of critical structures during minimally invasive spine surgery pushes the field by providing surgeons with detailed 3D visualizations of the surgical field without the need for large incisions. Studies have shown that the use of augmented reality (AR) technologies in minimally invasive spinal surgery has led to an average of 20% reduction in operative time, a 15% decrease in intraoperative blood loss, and faster recovery times compared to conventional methods [47]. These advancements have enhanced surgeons' ability to navigate small, complex anatomical spaces with greater accuracy, reducing the need for intraoperative fluoroscopy and exposure to radiation [47]. Furthermore, AR-assisted pedicle screw placement has significantly reduced the rate of misplaced screws, even when performed by less experienced surgeons, indicating a shorter learning curve for this complex procedure [47].

Postoperative Monitoring

Computer vision systems have emerged as promising tools for analyzing patient gait patterns after TKA. These systems utilize AI and image recognition techniques to provide objective and quantitative assessments of gait characteristics [8,51]. By capturing and analyzing video footage of patients walking, computer vision can detect subtle changes in gait parameters such as stride length, gait speed, and knee flexion angles [8]. This noninvasive approach offers several advantages over traditional gait analysis methods, including reduced setup time, increased patient comfort, and the ability to collect data in more natural settings. Recent studies have demonstrated the effectiveness of computer vision in identifying gait improvements following TKA, with patients showing significant enhancements in mobility and reduced pain within weeks of surgery [51]. Furthermore, these systems can aid in personalizing rehabilitation programs and monitoring patient progress over time, potentially leading to better long-term outcomes for TKA patients [8,51].

AI has also emerged as a tool for processing wearable sensor data to track rehabilitation progress. Algorithms can analyze complex data streams from various sensors, including accelerometers, gyroscopes, and force sensors, to provide comprehensive insights into patients' movement patterns and functional improvements [33,35]. ML techniques, such as supervised learning and DL models such as CNNs and long short-term memory, are particularly effective in automatically extracting features and recognizing temporal patterns in sensor data [35]. This allows for real-time monitoring of patients'

gait characteristics, joint mobility, and overall activity levels, enabling healthcare providers to make data-driven decisions about treatment plans. For instance, AI algorithms can detect subtle changes in gait parameters or upper limb function that may not be apparent through traditional clinical assessments. By continuously analyzing sensor data, these algorithms can also identify potential issues early on, allowing for timely interventions and personalized adjustments to rehabilitation programs [44]. The integration of AI and wearable sensors thus offers a promising approach to enhance the precision and effectiveness of rehabilitation monitoring, potentially leading to improved patient outcomes and more efficient healthcare delivery.

Automated detection of postoperative complications through imaging and clinical data analysis can enhance patient care and improve surgical outcomes. Advanced ML techniques, particularly DL models, have demonstrated significant potential in analyzing medical imaging data to predict surgical complexity and identify potential complications [7]. These models can process preoperative CT images to forecast surgical complexity with greater accuracy than expert surgeon judgment [7]. In addition, natural language processing algorithms have shown superior performance in identifying postoperative complications from electronic medical records compared to traditional patient safety indicators based on discharge coding [30]. These automated systems can detect a range of complications, including acute renal failure, deep vein thrombosis, pulmonary embolism, sepsis, pneumonia, and myocardial infarction, with higher sensitivity than conventional methods [30]. Furthermore, CNNs have been applied to analyze imaging data for bone and joint infection complications, offering improved resolution and more accurate lesion area recognition compared to traditional approaches [27]. The integration of these automated detection methods into clinical practice has the potential to significantly advance perioperative management and improve patient outcomes.

Clinical Examples

Several studies have explored advanced versus conventional surgical planning, focusing on outcomes and component positioning accuracy. One notable study with 60 patients compared AI-based 3D planning to traditional 2-dimensional (2D) X-ray planning for TKA; the AI group achieved significantly higher accuracy in predicting prosthesis size, valgus correction angle, and hip-knee-ankle angle [18]. In addition, patients in the AI group demonstrated better functional outcomes, as measured by the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) and the American Knee Society knee function scores at 3-, 6-, and 12-month post-surgery. Another study compared 60 TKAs performed using computer-assisted techniques with those using conventional methods [28]. The findings highlighted that computer-assisted TKA exhibited a significantly smaller range of alignment deviation in the frontal plane, specifically between 2.9° valgus to 3.1° varus, compared to the conventional approach, which ranged from 4.8° valgus to 6.6° varus. Moreover, the mean deviation from the mechanical axis was notably lower in the computer-assisted group at $1.4° \pm 0.9°$, compared to $2.0° \pm 1.7°$ in the conventional group.

In spine surgery, studies have demonstrated that AI models can optimize screw trajectories and dimensions based on preoperative imaging data, leading to improved accuracy compared to traditional freehand techniques. For instance, a study involving 208 pedicle screw placements showed that AI-assisted planning achieved an 85.1% classification as Gertzbein Grade A (no cortical breach), significantly outperforming the 64.9% accuracy of freehand placements [13]. Another innovative system, called Bone's Trajectory, utilized AI-based finite element analysis to identify optimal screw paths with maximal bone mineral density and pullout force, particularly beneficial for elderly osteoporotic patients [26]. Furthermore, robotic-assisted systems have demonstrated an impressive accuracy rate of approximately 95%, with studies indicating that the integration of AI in surgical navigation can reduce the learning curve for less experienced surgeons [2]. The use of AR in conjunction with AI has also shown promise, providing real-time visualization of anatomical structures, which may further enhance the safety and efficacy of pedicle screw placements [39]. Collectively, these advancements underscore the potential of AI-powered technologies in spinal surgery, paving the way for more standardized and safer surgical practices across diverse clinical settings.

AI-assisted THA has shown advantages over conventional methods in multiple studies. AI-assisted preoperative planning consistently outperforms traditional 2D X-ray template planning in prosthesis size prediction accuracy, with acetabular and femoral prosthesis prediction rates exceeding those of the traditional method across diverse patient populations [45,49]. Moreover, AI-assisted approaches result in more precise acetabular cup positioning, with higher percentages of placements within safe zones such as those defined by Lewinnek and Callanan [45,49]. AI techniques also significantly reduce lower-limb length discrepancies compared to traditional methods, enhancing postoperative limb symmetry [46]. Studies highlight improved operative efficiency with shorter surgical times and reduced intraoperative model testing in AI-assisted THA [49]. In terms of functional outcomes, patients receiving AI-guided procedures exhibit superior early postoperative Harris hip scores and faster recovery of hip function [49]. These findings collectively underscore AI potential in optimizing THA planning and execution, offering improved precision, efficiency, and patient outcomes.

Emerging Technologies: LLMs

LLMs have begun to make significant contributions to orthopedic surgery, in clinical practice and research. These advanced AI systems, built on transformer architectures, excel in natural language processing and are particularly suited for managing and interpreting the vast amounts of text-based data generated in medical practice [34]. Their ability to understand and generate coherent, contextually appropriate language makes them valuable tools in orthopedic surgery, where precise communication and documentation are essential.

In clinical settings, LLMs have shown utility in automating routine documentation tasks, such as generating clinical notes and patient letters [3]. By leveraging ambient documentation systems, LLMs can create standardized notes directly from patient-physician interactions, saving time and reducing the administrative burden on orthopedic surgeons [41]. This allows clinicians to focus more on patient care while ensuring that the documentation adheres to professional standards. In addition, LLMs are proving useful in-patient education, where AI-based chatbots can deliver clear explanations of complex orthopedic procedures, provide pre- and postoperative instructions, and answer frequently asked questions [10]. These tools enhance patient understanding and engagement, promoting better adherence to treatment plans. In research, LLMs are valuable assistants for orthopedic surgeons working on literature reviews and manuscript preparation. They can efficiently synthesize findings from large volumes of medical literature, identifying relevant studies and highlighting key insights [5]. Furthermore, their ability to generate concise, well-structured text aids researchers in articulating their findings and producing high-quality publications. Some models, like Llama, have even been fine-tuned with medical datasets, enabling more targeted applications in orthopedics [40].

The potential of multimodal LLMs, such as Google's Gemini, extends further into areas requiring the integration of text and image data, such as the interpretation of radiographs or MRI scans [38]. Such capabilities could complement the expertise of orthopedic surgeons in diagnosing and planning treatments. Similarly, Claude's analytical strengths could be harnessed for complex medical reasoning tasks in orthopedics, including treatment planning and case analysis [36]. Despite these advances, the limitations of LLMs must be carefully considered. They lack genuine medical understanding and are prone to hallucinations, the generation of plausible but incorrect information [6]. This necessitates rigorous verification of AI-generated outputs by qualified clinicians. Ethical considerations, such as maintaining transparency with patients about the use of AI, ensuring data privacy, and addressing biases in AI outputs, are paramount. While LLMs cannot replace the expertise and judgment of orthopedic surgeons, their integration into the field offers promising opportunities to enhance efficiency, patient education, and research productivity.

Emerging Technologies: Agentic AI

Agentic AI represents an advancement in orthopedic surgery, offering systems with the capability to perceive their environment, make decisions, and execute actions autonomously within predefined parameters [4,11,43]. These systems

leverage advanced technologies, such as reinforcement learning and neural networks, to interpret complex data and interact dynamically with clinical scenarios, making them valuable tools in modern surgical practice.

One of the most promising orthopedic applications of agentic AI is its potential integration into robotic-assisted surgery. These systems could analyze real-time intraoperative feedback, imaging data, and surgical plans to enhance precision and adaptability during procedures. For example, an agentic AI could adjust robotic movements during a total joint replacement to account for unexpected anatomical variations, improving outcomes and minimizing surgical errors. Similarly, these systems could play a pivotal role in preoperative planning by synthesizing imaging and patient-specific data to create optimized surgical blueprints. Agentic AI is also poised to revolutionize postoperative care. Autonomous systems could monitor patient recovery using wearable sensors and imaging, providing early detection of complications such as infection or prosthetic loosening. By continuously analyzing patient data, these systems can provide timely alerts and actionable recommendations to clinicians, ensuring proactive management and better long-term outcomes. Furthermore, agentic AI could support rehabilitation by tailoring physical therapy regimens based on real-time patient performance data, promoting faster recovery and personalized care.

However, the autonomy of these systems necessitates careful consideration of safety and accountability. In orthopedic surgery, where precision and trust are paramount, rigorous validation and robust error detection mechanisms are essential. Explainable AI techniques are critical to ensure transparency in decision-making, allowing surgeons to understand and validate AI recommendations before acting on them [31,32]. This is particularly important in high-stakes environments where patient safety is directly affected by AI-driven decisions. Ethical considerations are equally significant. Agentic AI systems must be designed to provide unbiased care across diverse patient populations. In orthopedics, this means accounting for variations in anatomy, demographics, and comorbidities to deliver equitable and effective solutions. Continuous monitoring, iterative updates, and recalibrations are vital to maintaining performance and adapting to new clinical challenges.

By combining situational awareness, precision, and adaptability, agentic AI has the potential to enhance but not replace the capabilities of orthopedic surgeons, potentially serving as valuable collaborators in care delivery. As technology evolves, orthopedic surgery stands to benefit significantly from the thoughtful integration of agentic AI, provided safety, transparency, and ethical standards are rigorously upheld.

Conclusion

AI is rapidly transforming orthopedic surgery through diverse applications spanning diagnostic assistance, surgical planning, and clinical documentation. While ML, DL, and LLMs demonstrate remarkable potential in enhancing surgical precision and efficiency, these technologies are designed to augment rather than replace clinical expertise. The successful implementation of AI requires careful attention to validation, regulatory compliance, and healthcare system integration while addressing challenges in data quality and standardization. As emerging technologies such as agentic AI and autonomous systems continue to evolve, maintaining the balance between innovation and patient safety remains crucial, with successful outcomes dependent on proper validation, risk management, and sustained human oversight in clinical decision-making. The future of AI in orthopedics points toward increasingly personalized treatment approaches and streamlined workflows, provided these advances continue to prioritize patient care and healthcare equity.

Supplemental Material

Supplemental material, sj-docx-1-hss-10.1177_15563316251339596 for Artificial Intelligence and Musculoskeletal Surgical Applications by Felix C. Oettl, Bálint Zsidai, Jacob F. Oeding and Kristian Samuelsson in HSS Journal®

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Footnotes

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

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